

Application Title: Vocalis

What does the application do:

The application is used to help people improve their communication.

The application has access to camera and microphone. It majorly focuses on the following things that are used as evaluation metrics for the speaker;

Scoring Criteria:

Dimension	Parameter	What system measures	Example metric	Scoring direction
Eye Contact	Camera gaze	% time looking toward camera	Camera-facing %	Higher → better
	Gaze-away frequency	How often gaze leaves camera	Events/min	Lower → better
	Gaze-away duration	Length of each gaze break	Avg sec	Lower → better
	Gaze stability	Stability of camera-facing gaze	Variance	Moderate → better
	Downward gaze	Looking at notes/screen	% duration	Lower → better
Posture	Upright posture	Spine/body alignment	% duration upright	Higher → better
	Slouching	Forward/downward posture	% duration	Lower → better
	Leaning	Excessive forward/backward lean	Avg deviation	Lower → better
	Shoulder alignment	Shoulder symmetry/alignment	Alignment score	Higher → better
	Body stability	Unnecessary body movement	Movement variance	Moderate → better
Hand Gestures	Gesture frequency	Number of hand gestures	Gestures/min	Moderate → better
	Gesture amplitude	Size of gestures	Movement range	Context-dependent
	Gesture duration	How long gestures persist	Avg sec	Context-dependent
	Gesture-speech sync	Gesture alignment with speech	Sync %	Higher → better
	Repetitive gestures	Same gesture repeated	Events/min	Lower → better
	Hand visibility	Hands visible in frame	% duration	Higher → better
Body Movement	Body sway	Side-to-side movement	Sway frequency	Lower → better
	Rocking	Repetitive forward/back movement	Events/min	Lower → better

	Fidgeting	Small repetitive movements	Events/min	Lower → better
	Movement intensity	Overall movement level	Movement index	Moderate → better
Facial Expression	Expression variation	Range of facial expressions	Variation index	Moderate → better
	Smile	Smile occurrence	% duration	Context-dependent
	Neutral expression	Extended neutral face	% duration	Context-dependent
	Eyebrow movement	Facial emphasis	Events/min	Context-dependent
	Expression-speech alignment	Expression matches delivery	Alignment score	Higher → better
Head Movement	Head stability	Excessive head movement	Movement variance	Moderate → better
	Nodding	Nodding frequency	Events/min	Context-dependent
	Head orientation	Facing camera	% duration	Higher → better
	Head tilt	Excessive tilt	Avg deviation	Lower → better
Voice — Rate	Speech rate	Speaking speed	Words/min	Optimal range
	Rate variation	Pace changes	WPM variance	Moderate → better
Voice — Pauses	Pause frequency	Number of pauses	Pauses/min	Context-dependent
	Pause duration	Average pause length	Avg sec	Optimal range
	Long pauses	Unusually long silence	Count	Lower → better
	Pause placement	Pauses at natural points	Placement score	Higher → better
Voice — Fillers	Filler frequency	"um", "uh", "like", etc.	Fillers/min	Lower → better
	Filler ratio	Fillers relative to speech	Fillers/total words	Lower → better
Voice — Volume	Average volume	Overall loudness	dB/LUFS	Optimal range
	Volume variation	Loudness variation	SD	Moderate → better
	Volume stability	Sudden loud/quiet changes	Variation index	Higher → better
Voice — Pitch	Mean pitch	Average fundamental frequency	F0 Hz	Context-dependent
	Pitch range	Vocal pitch range	F0 range	Moderate → better
	Pitch variation	Pitch dynamics	F0 SD	Moderate → better

	Monotony	Low pitch variation	Monotony index	Lower → better
Voice — Delivery Energy	Vocal energy	Voice intensity	RMS energy	Moderate → better
	Energy variation	Dynamic delivery	Energy SD	Moderate → better
Voice — Articulation	Speech clarity	Clarity of spoken words	Clarity score	Higher → better
	Mumbled speech	Reduced articulation	% unclear speech	Lower → better
	Articulation rate	Syllables during phonation	Syllables/sec	Optimal range
Voice — Pronunciation	Pronunciation accuracy	Word pronunciation	Error rate	Lower → better
	Pronunciation consistency	Same word pronounced consistently	Consistency %	Higher → better
Voice — Delivery	Sentence completion	Complete vs trailing sentences	Completion %	Higher → better
	Trailing-off	Voice fading at sentence end	Events	Lower → better
	Repetition	Repeated words/phrases	Repetitions/min	Lower → better
	Self-correction	Restarting/correcting speech	Events/min	Lower → better
Voice — Breathing	Breath pauses	Pauses caused by breathing	Events/min	Context-dependent
	Breath frequency	Breathing interruptions	Events/min	Context-dependent
Voice — Stability	Pitch stability	Sudden pitch fluctuations	Variation index	Moderate → better
	Volume stability	Sudden loudness changes	Variation index	Moderate → better
	Vocal break	Voice instability/breaks	Events	Lower → better
Silence	Total silence	Total non-speaking time	Silence %	Context-dependent
	Silence distribution	How silence is distributed	Distribution index	Moderate → better
Content	Introduction	Opening clarity	Intro score	Higher → better
	Structure	Logical flow	Structure score	Higher → better
	Relevance	Topic alignment	Relevance score	Higher → better
	Repetition	Repeated ideas	Repetition ratio	Lower → better
	Key-point coverage	Important points addressed	Coverage %	Higher → better
	Conclusion	Ending clarity	Conclusion score	Higher → better

Presentation Quality	Camera framing	Face/body positioning	Framing score	Higher → better
	Face visibility	Face unobstructed	% duration	Higher → better
	Lighting	Face illumination	Lighting score	Higher → better
	Background noise	Non-speech noise	dB/noise ratio	Lower → better
	Audio quality	Recording quality	SNR/clipping	Higher → better
	Camera stability	Camera movement	Motion score	Higher → better